

확률적 계획법 HW 1

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Problem 1.1 (Convex Sets)

Determine whether each of the following statement is True or False. In either case, provide a proof or a counterexample.

- (a) All polyhedral sets are convex.
- (b) The intersection of convex sets is convex.
- (c) The union of convex sets is convex.

(a) True.

A set $C \subseteq \mathbb{R}^n$ is a convex set if for all $x_1, x_2 \in C$,
we have $\lambda x_1 + (1-\lambda)x_2 \in C$, $\forall \lambda \in [0, 1]$

A set $C \subseteq \mathbb{R}^n$ is polyhedral if it can be written as
 $C = \{x \in \mathbb{R}^n \mid Ax \geq b\}$ for some matrix $A \in \mathbb{R}^{m \times n}$
and a vector $b \in \mathbb{R}^m$.

Let set C_1 be a polyhedral set. Namely, $C_1 = \{x \in \mathbb{R}^n \mid A_1 x \geq b_1\}$ (where $A_1 \in \mathbb{R}^{m_1 \times n}$, $b_1 \in \mathbb{R}^{m_1}$).
Pick arbitrary x_1, x_2 from C_1 . Then, $A_1 x_1 \geq b_1$ and $A_1 x_2 \geq b_1$ holds.

Then, for $\forall \lambda \in [0, 1]$,

$$A_1 x_1 \geq b_1 \Rightarrow A_1 (\lambda x_1) \geq \lambda b_1$$

$$A_1 x_2 \geq b_1 \Rightarrow A_1 \{(1-\lambda)x_2\} \geq (1-\lambda)b_1$$

$$\Rightarrow A_1 \{\lambda x_1 + (1-\lambda)x_2\} \geq b_1$$

$$\Rightarrow \lambda x_1 + (1-\lambda)x_2 \in C_1$$

Therefore, C_1 is convex.

Therefore, All polyhedral sets are convex. ■

(b) True.

C_1 과 C_2 가 모두 convex set 이라고 하자. Pick two arbitrary points $x_1, x_2 \in C_1 \cap C_2$.

Then, for $\forall \lambda \in [0, 1]$, $\lambda x_1 + (1-\lambda)x_2 \in C_1$ and

$\lambda x_1 + (1-\lambda)x_2 \in C_2$ 가 성립한다.

$$\Rightarrow \lambda x_1 + (1-\lambda)x_2 \in C_1 \cap C_2. \Rightarrow C_1 \cap C_2 \text{ is convex set.} \blacksquare$$

(c) False.

반례: Let $C_1 = [0, 1] \subseteq \mathbb{R}$, $C_2 = [2, 3] \subseteq \mathbb{R}$. 그러면, C_1, C_2 는 각각 convex set이다.

그런데, $1 \in C_1 \cap C_2$, $2 \in C_1 \cap C_2$. 그러면, $\frac{1}{2}x_1 + (1-\frac{1}{2})x_2 = \frac{1}{2} + 1 = \frac{3}{2} \notin C_1 \cap C_2$.

$\Rightarrow C_1 \cup C_2$ 는 convex set이 아님. ■

Problem 1.5 (Epigraph)

Determine whether each of the following statement is True or False. In either case, provide a proof or a counterexample.

- (a) The epigraph of a concave function is a convex set.
- (b) The epigraph of a convex function is a convex set.

(a) False.

예시: Let $f(x) = -x^2 + 1$. $f(x)$ is concave function.

$$(-1, 0) \in \text{epi-}f, \quad (1, 0) \in \text{epi-}f.$$

$$\text{그런데, } \frac{1}{2}(-1, 0) + (1-\frac{1}{2})(1, 0) = (0, 0) \notin \text{epi-}f. \quad (\because 0 < f(0) = 1).$$

$\Rightarrow \text{epi-}f$ is not a convex set. ■

(b) True.

function $f: C \rightarrow \mathbb{R}$ 일 때, $(\text{epi-}f, C \subseteq \mathbb{R}^n)$.

$$\text{epi-}f := \{(x, \alpha) \in \mathbb{R}^{n+1} : \alpha \geq f(x)\} \quad \text{이다.}$$

Consider $(x, f(x)) \in \mathbb{R}^{n+1}$.

If f is convex, then for any $x_1, x_2 \in C$, $\forall \lambda \in [0, 1]$,
$$\lambda(x_1, f(x_1)) + (1-\lambda)(x_2, f(x_2)) = (\lambda x_1 + (1-\lambda)x_2, \lambda f(x_1) + (1-\lambda)f(x_2))$$
$$\geq (\lambda x_1 + (1-\lambda)x_2, f(\lambda x_1 + (1-\lambda)x_2))$$

Let $(x_1, \alpha_1), (x_2, \alpha_2) \in \text{epi-}f$ such that $\alpha_1 \geq f(x_1), \alpha_2 \geq f(x_2)$.

$$\begin{aligned} \text{Then, } & \lambda(x_1, \alpha_1) + (1-\lambda)(x_2, \alpha_2) \\ &= (\lambda x_1 + (1-\lambda)x_2, \lambda \alpha_1 + (1-\lambda)\alpha_2) \\ &\geq (\lambda x_1 + (1-\lambda)x_2, \lambda f(x_1) + (1-\lambda)f(x_2)) \\ &\geq (\lambda x_1 + (1-\lambda)x_2, f(\lambda x_1 + (1-\lambda)x_2)) \end{aligned}$$

Therefore, $\lambda(x_1, \alpha_1) + (1-\lambda)(x_2, \alpha_2) \in \text{epi-}f$.

Therefore, epigraph of a convex function is a convex function. ■

Problem 1.6 (Function Properties)

Determine whether each of the following statement is True or False. In either case, provide a proof or a counterexample.

- (a) An affine function is both convex and concave. (Hint: consider an affine function given as $f(x) = \beta^T x + \alpha$.)
- (b) The function $f(x) = \max \{ \alpha_t + \beta_t^T x \mid t = 0, 1, \dots, k \}$ is concave.

(a) True.

Consider an affine function given as $f(x) = \beta^T x + \alpha$. (여기, β 는 n-vector, α 는 scalar).

해당 affine function의 정의역에서 임의의 원소 x, y 를 뽑자. Then, for $\forall \lambda \in [0, 1]$,

$$\begin{aligned} f(\lambda x + (1-\lambda)y) &= \beta^T (\lambda x + (1-\lambda)y) + \alpha = \lambda \beta^T x + (1-\lambda) \beta^T y + (1-\lambda + \lambda) \alpha \\ &= \lambda (\beta^T x + \alpha) + (1-\lambda) (\beta^T y + \alpha) \end{aligned}$$

$$\Rightarrow f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y) \text{ 도 성립하고,}$$

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y) \text{ 도 성립한다.}$$

Therefore, affine function is always convex and concave. ■

(b) False. 함수 $f(x)$ 가 concave하지가 아닌, $f(x)$ 정의역에 있는 임의의 x_1, x_2 에 대해,
 $f(\lambda x_1 + (1-\lambda)x_2) \geq \lambda f(x_1) + (1-\lambda)f(x_2)$, $\forall \lambda \in [0, 1]$ 이어야 한다.

다음과 같은 반례가 있다.

$$\text{Let } f(x) = \max \{ \alpha_t + \beta_t^T x \mid t = 0, 1 \}$$

$$\alpha_0 = 0 \quad \beta_0 = 1 \quad (\alpha_0 + \beta_0^T x = x)$$

$$\alpha_1 = 1 \quad \beta_1 = -1 \quad (\alpha_1 + \beta_1^T x = 1 - x)$$

$$\text{Let } x_1 = 0 \quad x_2 = 1$$

$$\text{Let } \lambda = 0.5$$

그러면,

$$f(\lambda x_1 + (1-\lambda)x_2) \geq \lambda f(x_1) + (1-\lambda)f(x_2) \text{ 이 성립하지 않음.}$$

$$(\because) f(0 + \frac{1}{2}) = \frac{1}{2} < \frac{1}{2}(1) + \frac{1}{2}(1) = 1$$

$\Rightarrow$ Concave하지 않음. ■

(c) The function $f(x) = \text{Max} \{q^T y \mid Wy \leq r - Tx, y \geq 0\}$ is convex. (Hint: assume that this LP is dual feasible and consider the dual problem of $\text{Max} \{q^T y \mid Wy \leq r - Tx, y \geq 0\}$.)

(d) The function $f(x) = \text{Min} \{q^T y \mid Wy \geq r - Tx, y \geq 0\}$ is convex. (Hint: assume that this LP is dual feasible and consider the dual problem of $\text{Min} \{q^T y \mid Wy \geq r - Tx, y \geq 0\}$.)

(c) False. 반례: Maximize y
s.t. $y \leq x,$
 $y \leq 2-x,$
 $y \geq 0.$

$x=0$ 일 경우: Maximize y
s.t. $y \leq 0,$
 $y \leq 2,$
 $y \geq 0.$

$$\Rightarrow f(0) = 0$$

$x=2$ 일 경우: Maximize y
s.t. $y \leq 2$
 $y \leq 0$
 $y \geq 0$

$$\Rightarrow f(2) = 0$$

$x=1$ 일 경우: Maximize y
s.t. $y \leq 1$
 $y \leq 1$
 $y \geq 0$
 $\Rightarrow f(1) = 1$

$\geq,$
 $f(\frac{1}{2}x_0 + (1-\frac{1}{2})x_2) = 1$
 $\geq \frac{1}{2}f(0) + (1-\frac{1}{2})f(2) = 0$
 $\therefore f(x)$ 는 convex 하지 않음. ■

(d)는 다음 장에.

(d) True.

무선, 해당 명제를 다음과 같이 다시 쓸 수 있다.

The function $f(b) = \min \{ C^T x \mid Ax \leq b, x \geq 0 \}$ is convex.

이는 참이다. 왜냐하면,

우선 $\min \{ C^T x \mid Ax \leq b, x \geq 0 \}$ 의 dual problem을 다음과 같이 쓸 수 있다.

$$\max \{ b^T y \mid A^T y \geq C, y \geq 0 \}.$$

b에 b_1 을 할당하면, optimal primal solution x_1 을 구할 수 있고,

optimal dual solution y_1 을 구할 수 있다.

By strong duality, $C^T x_1 = b_1^T y_1$ 이다.

이번엔 b에 $\lambda b_1 + (1-\lambda) b_2$ (단, $\lambda \in [0, 1]$)을 할당하라. 이때의 optimal primal solution을 x_λ 라고 하자.

또, 이때의 optimal dual solution을 y_λ 라고 하자.

그러면,

$$f(\lambda b_1 + (1-\lambda) b_2) = C^T x_\lambda$$

$$= (\lambda b_1 + (1-\lambda) b_2)^T y_\lambda \quad \dots \text{ (by strong duality)}$$

$$= \lambda b_1^T y_\lambda + (1-\lambda) b_2^T y_\lambda$$

$$\leq \lambda b_1^T y_1 + (1-\lambda) b_2^T y_2 \quad \dots \text{ (dual problem은 maximization problem.)}$$

$\dots$ (b에 b_2 를 할당 했을때

optimal primal solution을 x_2 ,

optimal dual solution을 y_2 라 하자.)

$$= \lambda C^T x_1 + (1-\lambda) C^T x_2 \quad \dots \text{ (by strong duality)}$$

$$= \lambda f(b_1) + (1-\lambda) f(b_2)$$

Therefore, for all b_1, b_2 , $\forall \lambda \in [0, 1]$,

$$f(\lambda b_1 + (1-\lambda) b_2) \leq \lambda f(b_1) + (1-\lambda) f(b_2)$$

$\Rightarrow f$ 는 convex function 이다. ■

(사실 유사한 방식으로 c 의 함수가 concave function임을 증명할 수 있다.)

Problem 1.9 (Random Variable)

Consider an experiment involving two tosses of a fair coin, where the outcome for each toss is either heads (H) or tails (T). The outcomes (first toss and second toss) are ordered pairs, e.g., HH means heads followed by heads, and TH means tails followed by heads. Answer the following questions:

- (a) What is the sample space?
- (b) How many elements are in the σ -algebra $\mathcal{A}$?
- (c) Write out explicitly the σ -algebra $\mathcal{A}$.
- (d) What is the probability measure $\mathbb{P}(\{\})$?

$$(a) \Omega = \{ HH, HT, TH, TT \}$$

$$(b) |\mathcal{A}| = 2^{|\Omega|} = 2^4 = 16$$

$$(c) \mathcal{A} = \{ \emptyset, \{HH\}, \{HT\}, \{TH\}, \{TT\}, \{HH, HT\}, \{HH, TH\}, \{HH, TT\}, \{HT, TH\}, \{HT, TT\}, \{TH, TT\}, \{HH, HT, TH\}, \{HH, HT, TT\}, \{HT, TH, TT\}, \{HH, HT, TH, TT\} \}$$

$$(d) \mathbb{P}(\{\}) = 0$$

$$\mathbb{P}(\{HH\}) = \mathbb{P}(\{HT\}) = \mathbb{P}(\{TH\}) = \mathbb{P}(\{TT\}) = \frac{1}{4}$$

$$\mathbb{P}(\{HH, HT\}) = \mathbb{P}(\{HH, TH\}) = \mathbb{P}(\{HH, TT\}) = \mathbb{P}(\{HT, TH\}) = \mathbb{P}(\{HT, TT\}) = \mathbb{P}(\{TH, TT\}) = \frac{1}{2}$$

$$\mathbb{P}(\{HH, HT, TH\}) = \mathbb{P}(\{HH, HT, TT\}) = \mathbb{P}(\{HT, TH, TT\}) = \frac{3}{4}$$

$$\mathbb{P}(\{HH, HT, TH, TT\}) = 1$$

Problem 1.10 (Multistage Model)

Consider a multistage SP with recourse and planning horizon $H = 3$ (three stages). Let the t -stage problem data be denoted $c^t(\omega_t) \in \mathbb{R}^{n_t}$, $T_t(\omega_t) \in \mathbb{R}^{m_t \times n_t}$, $W_t(\omega_t) \in \mathbb{R}^{m_t \times n_t}$, $r_t(\omega_t) \in \mathbb{R}^{m_t}$ for $t = 1, 2, 3$. Answer the following questions:

- Write the t -stage constraints $X_t(x_{t-1}, \omega_{[t]})$ for $t = 1, 2, 3$ explicitly and state any assumptions you wish to make.
- Using your definition of $X_t(x_{t-1}, \omega_{[t]})$ for $t = 1, 2, 3$ in part (a) write the Bellman equations for all the three stages.
- Write the explicit extensive form or DEP formulation.

(a) Assumptions 1) Nonanticipativity

2) Stage 1 decision is data independent

3) Data is stage wise independent

4) For each stage, there is at least one feasible solution $x_t \geq 0$,

assuming x_{t-1} is feasible

5) R.V. $\tilde{\omega}$ is discretely distributed with finite scenarios

6) Relatively complete recourse

$$(t=1) \quad X_1 = \{x_1 \in \mathbb{R}_+^{n_1} \mid T_1 x_1 \geq r_1\}$$

$$(t=2) \quad X_2(x_1, \omega_{[2]}) = \{x_2 \in \mathbb{R}_+^{n_2} \mid T_2(\omega_2) x_2 + W_2(\omega_2) x_1 \geq r_2(\omega_2)\}$$

$$(t=3) \quad X_3(x_2, \omega_{[3]}) = \{x_3 \in \mathbb{R}_+^{n_3} \mid T_3(\omega_3) x_3 + W_3(\omega_3) x_2 \geq r_3(\omega_3)\}$$

$$(b) \quad (t=1) \quad \min_{x_1 \in X_1} C_1^T x_1 + E[f_2(x_1, \omega_2)]$$

$$(t=2) \quad \min_{x_2(\omega_2) \in X_2(x_1, \omega_{[2]})} C_2^T(\omega_2) x_2 + E[f_3(x_2, \omega_{[3]}) \mid \omega_{[2]}]$$

$$(t=3) \quad \min_{x_3(\omega_3) \in X_3(x_2, \omega_{[3]})} C_3^T(\omega_3) x_3$$

(c) Let x_{s_t} be the decision variable for scenario $s_t \in S_t$ (sample space)

$$\min C_1^T x_1 + \sum_{s_2 \in S_2} P_{s_2} C_2^T(\omega_{s_2}) x_{s_2} + \sum_{s_3 \in S_3} P_{s_3} C_3^T(\omega_{s_3}) x_{s_3}$$

$$\text{s.t. } T_1 x_1 \geq r_1,$$

$$T_2^{s_2} x_2^{s_2} + W_2^{s_2} x_1 \geq r_2^{s_2}, \quad \forall s_2 \in S_2$$

$$T_3^{s_3} x_3^{s_3} + W_3^{s_3} x_2 \geq r_3^{s_3}, \quad \forall s_3 \in S_3$$

$$x_1, x_2, x_3 \geq 0$$

$$P_{s_t} \geq 0, \quad \forall s_t$$

$$\sum_{s_t \in S_t} P_{s_t} = 1$$